

CHERISH VASANT

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PROFILE

I'm a final-year CSE student, interested in AI, ML, Deep Learning, NLP, Generative AI, and Agentic AI systems. I enjoy understanding how these models work and how they can be used to solve real-world problems. I love problem solving, and I learn best by building projects and experimenting with ideas. I'm always trying to go deeper, improve my skills, and turn what I learn into something practical.

EDUCATION

Vellore Institute of Technology

B.Tech in Computer Science and Engineering (spec. AI & ML) – CGPA: 9.24/10.0

Chennai, India

Aug 2023 – May 2027

Sishya OMR

Grade 12 – ISC, Science with Computer Science – 95.8%

Chennai, India

2021 – 2023

St. Patrick's High School

Grade 10 – CISCE – 94.8%

Chennai, India

2019 – 2021

EXPERIENCE

Linaius

AI Intern

Remote

May 2026 – Jun 2026

- Learned how RAG systems work, BM25-based retrieval and benchmark-driven evaluation practices, alongside NLP text-segmentation techniques like TextTiling for large-scale language pipelines.
- Designed prompting strategies (task decomposition, two-stage extraction, neutral query framing) and made programmatic LLM calls through AWS Bedrock, with cost tracking and performance optimizations including caching, streaming, and concurrent execution.

Indian Space Academy (ISA)

Research Intern

Remote

Jun 2025 – Jul 2025

- Developed the **Age of the Universe Estimation** project using the Pantheon+SH0ES Type Ia Supernova dataset, fitting a flat Λ CDM cosmological model to estimate the Hubble constant ($H_0 = 72.97$ km/s/Mpc) and the age of the universe (≈ 12.36 Gyr) via numerical integration.
- Worked extensively with astronomical datasets, FITS files, and the Astropy ecosystem for scientific data processing and analysis in Python.

FlyBrand Digital Solutions

AI Content Intern

Remote

May 2025 – Jun 2025

- Explored AI model development workflows using Google Teachable Machine and Roboflow, building image, audio, and computer vision models for posture recognition, voice-command classification, and PCB defect detection.
- Produced technical training demonstrations documenting model creation, dataset preparation, training, evaluation, and deployment workflows for educational AI content.

PROJECTS

TrebleAI

AI-Powered Music Learning Platform, LangChain Agents, Full-Stack

- Designed and built **TrebleAI**, a self-learning music platform enabling users to independently learn and practice sheet music, powered by an end-to-end pipeline (OpenCV/PyMuPDF, Audiveris OMR, music21, FluidSynth) converting sheet music into synchronized audio and interactive notation.

- Built LangChain-based tutoring agents (a practice agent with context of the uploaded piece, and a music-theory tutor) with structured tool-calling, integrated into a full-stack application (FastAPI, Next.js, PostgreSQL) featuring synchronized sheet-music rendering, animated piano visualization, and real-time playback.

Cloud Security for Remote Patient Monitoring Systems

Machine Learning, Cybersecurity, IoT (Python, scikit-learn, XGBoost, CatBoost)

- Developed a multi-class intrusion detection system on the **Edge-IIoTset** dataset (157,800 records) to classify normal traffic against **14 cyberattack categories** (DDoS, MITM, SQL injection, ransomware, port scanning), using a feature engineering and RFE-based selection pipeline (65 to 40 features) with SMOTE to correct severe class imbalance.
- Benchmarked 8 classifiers spanning tree, boosting, and ensemble methods, achieving **99.02% test accuracy** with a Stacking ensemble (Random Forest + XGBoost + CatBoost base estimators, Logistic Regression), with single-estimator XGBoost close behind at 99.01% for a fraction of the model size.

Atrial Fibrillation Detection from ECG Signals

Deep Learning, Time Series, Biomedical AI, Team Project (PyTorch, TCN, EfficientNet)

- Developed a hierarchical deep learning pipeline on the **PhysioNet/CinC 2017 Challenge** dataset (8,528 ECG recordings) to classify Normal, Atrial Fibrillation, Other Arrhythmia, and Noisy signals under significant class imbalance, using a two-stage architecture: a Dilated TCN-ResNet + feature-MLP gate (Stage 1) feeding an EfficientNet-B2 spectrogram classifier (Stage 2).
- Achieved **85.98% end-to-end test accuracy** and **0.7720 macro F1-score**, outperforming MLP, Random Forest, Gradient Boosting, and XGBoost baselines (61–67% macro F1).

Chronic Kidney Disease Prediction

Machine Learning, Healthcare AI, Flask, Team Project (XGBoost, scikit-learn)

- Developed an ML pipeline for early CKD prediction on the UCI CKD dataset, applying missing-value analysis, clinical range validation, and feature engineering (eGFR, comorbidity scores), then benchmarked and hyperparameter-tuned classifiers across baseline, feature-engineered, and feature-selected dataset versions.
- Rejected configurations scoring a suspicious 100% test accuracy due to proxy-variable leakage risk, selecting the best-generalizing configuration at **98.75% test accuracy** (98.00%/100% recall) and integrating it into a Flask web app for real-time clinical decision support.

PATENT

Automatic Speech Recognition and Translation System for the Toda Language leveraging Pretrained Models

Patent Application Filed

- Building an automatic speech recognition and translation system for endangered, low-resource languages, with an initial focus on the Toda language; the approach generalizes to other low-resource languages. Patent application submitted for filing.

SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript, HTML/CSS

AI & Machine Learning: PyTorch, TensorFlow/Keras, Scikit-learn, XGBoost, CatBoost, Deep Learning, Computer Vision, NLP, LLMs, LangChain, Prompt Engineering, Pandas, NumPy, Matplotlib, OpenCV

Databases & Tools: PostgreSQL, Git, VS Code, Cursor, Claude Code, AWS Bedrock

COURSES

Machine Learning – SmartBridge, in collaboration with Google for Developers (May – Jun 2025)

C Programming – Spoken Tutorial Project, IIT Bombay – Score: 90% (Apr 2024)

C++ Programming – Spoken Tutorial Project, IIT Bombay – Score: 80% (Apr 2024)